Satisfaction

```
sat <- read.csv("satisfaction.csv")
```

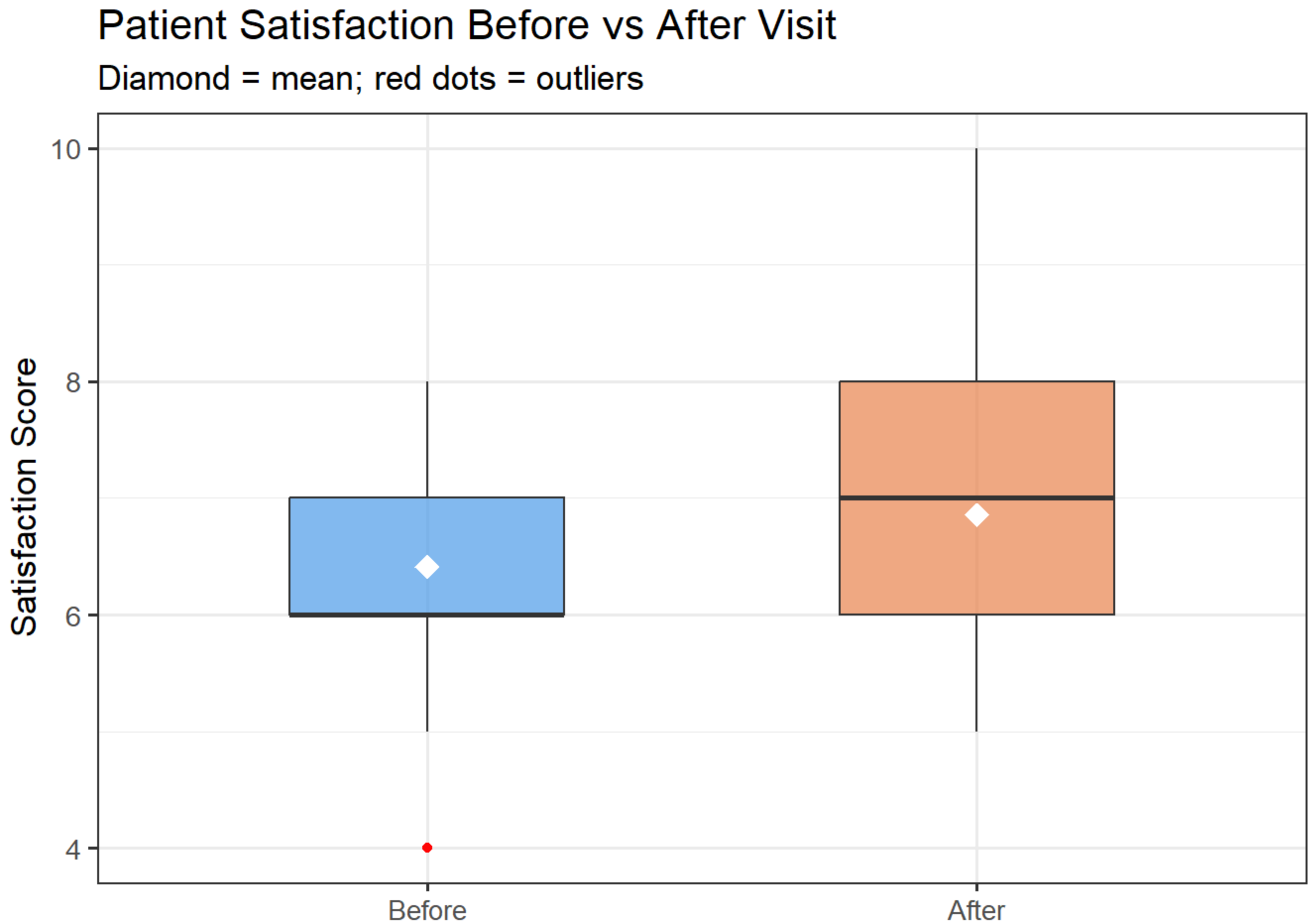
```
library(ggplot2)
```

```
sat_long <- data.frame(
  Score  = c(sat$sat.before, sat$sat.after),
  Timing = rep(c("Before", "After"), each = nrow(sat))
)
sat_long$Timing <- factor(sat_long$Timing, levels = c("Before", "After"))

ggplot(sat_long, aes(x = Timing, y = Score, fill = Timing)) +
  geom_boxplot(outlier.colour = "red", width = 0.5, alpha = 0.7) +
  stat_summary(fun = mean, geom = "point", shape = 18, size = 4, colour = "white") +
  scale_fill_manual(values = c("Before" = "#4C9BE8", "After" = "#E8834C")) +
  labs(
    title    = "Patient Satisfaction Before vs After Visit",
    subtitle = "Diamond = mean; red dots = outliers",
    x        = NULL,
    y        = "Satisfaction Score"
  ) +
  theme_bw(base_size = 13) +
  theme(legend.position = "none")
```

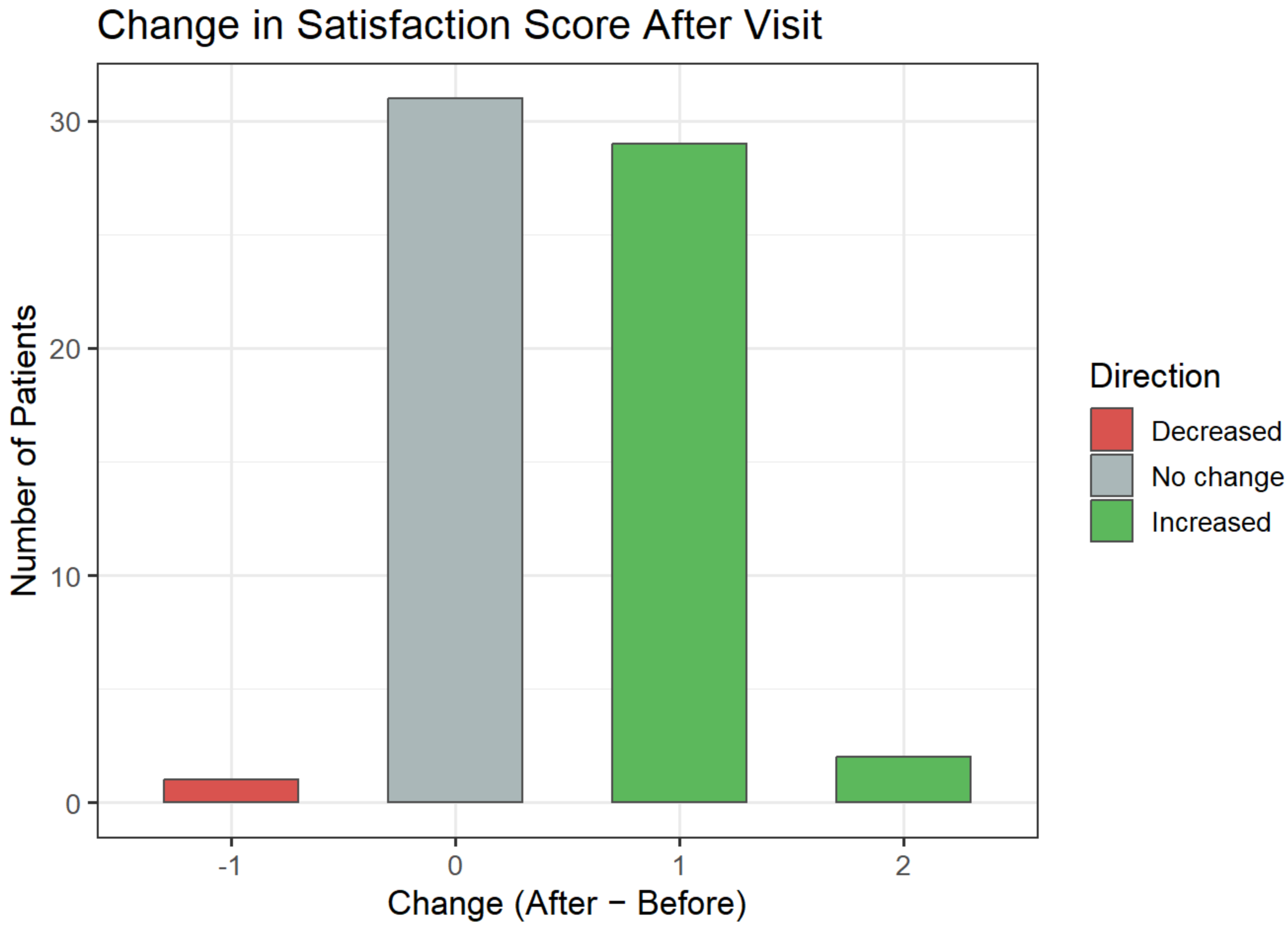
```
## Warning: Removed 5 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
## Warning: Removed 5 rows containing non-finite outside the scale range
## (`stat_summary()`).
```



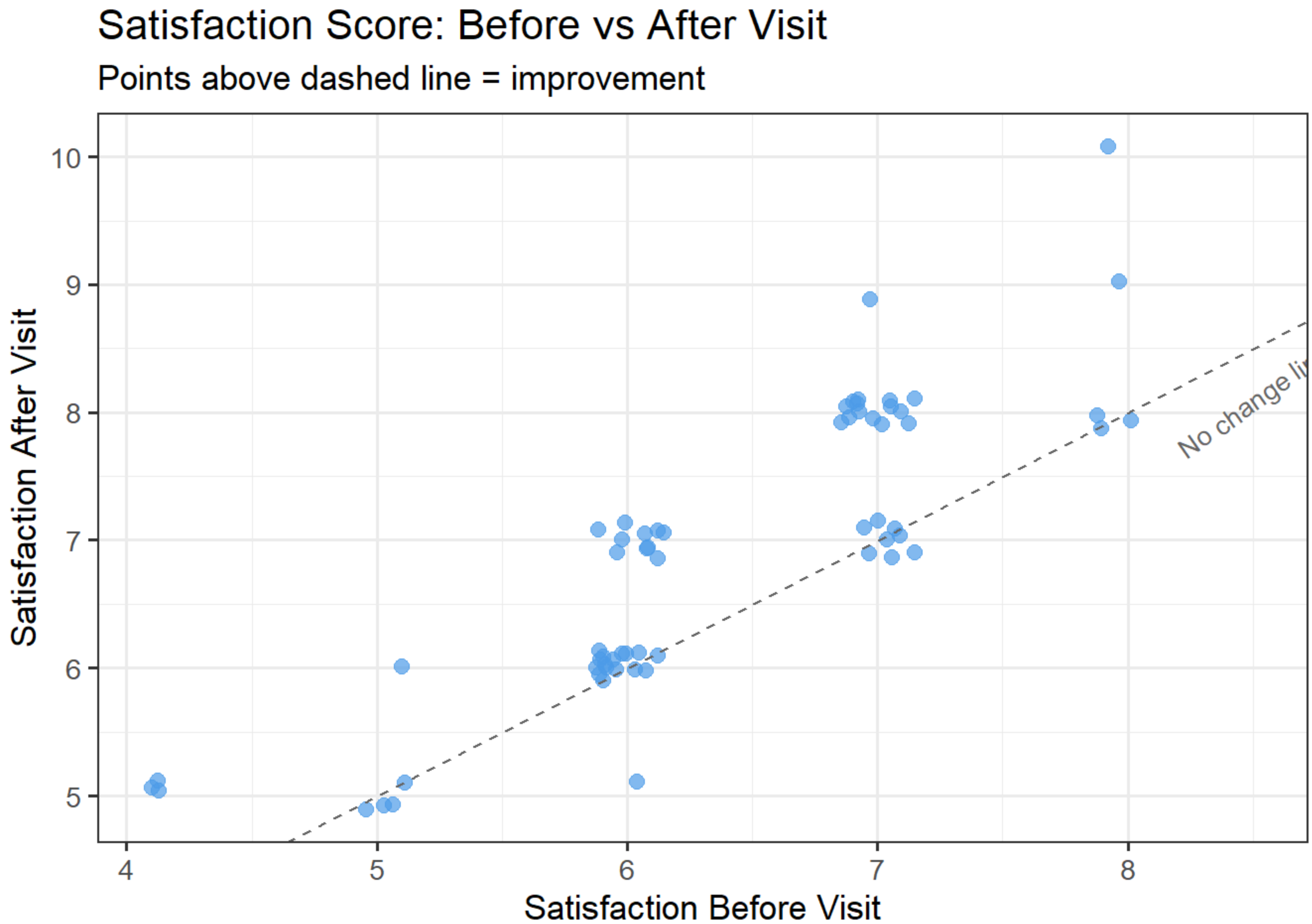

```
change_counts <- as.data.frame(table(Change = sat$change))
change_counts$Change <- as.numeric(as.character(change_counts$Change))

ggplot(change_counts, aes(x = factor(Change), y = Freq,
                           fill = factor(sign(Change),
                                         labels = c("Decreased", "No change", "Increased")))) +
  geom_col(colour = "grey30", width = 0.6) +
  scale_fill_manual(
    values = c("Decreased" = "#D9534F", "No change" = "#AAB7B8", "Increased" = "#5CB85C"),
    name   = "Direction"
  ) +
  labs(
    title = "Change in Satisfaction Score After Visit",
    x     = "Change (After − Before)",
    y     = "Number of Patients"
  ) +
  theme_bw(base_size = 13)
```




```
ggplot(sat, aes(x = sat.before, y = sat.after)) +
  geom_jitter(width = 0.15, height = 0.15,
              colour = "#4C9BE8", alpha = 0.7, size = 2.5) +
  geom_abline(intercept = 0, slope = 1,
              linetype = "dashed", colour = "grey40") +
  annotate("text", x = 8.5, y = 8.1, label = "No change line",
           colour = "grey40", size = 3.5, angle = 35) +
  scale_x_continuous(breaks = 4:10) +
  scale_y_continuous(breaks = 4:10) +
  labs(
    title    = "Satisfaction Score: Before vs After Visit",
    subtitle = "Points above dashed line = improvement",
    x        = "Satisfaction Before Visit",
    y        = "Satisfaction After Visit"
  ) +
  theme_bw(base_size = 13)
```

```
## Warning: Removed 5 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Satisfaction Score: Before vs After Visit
Points above dashed line = improvement
10
9
8
7
6
5
4
5
6
7
8
Satisfaction After Visit
Satisfaction Before Visit
No change li